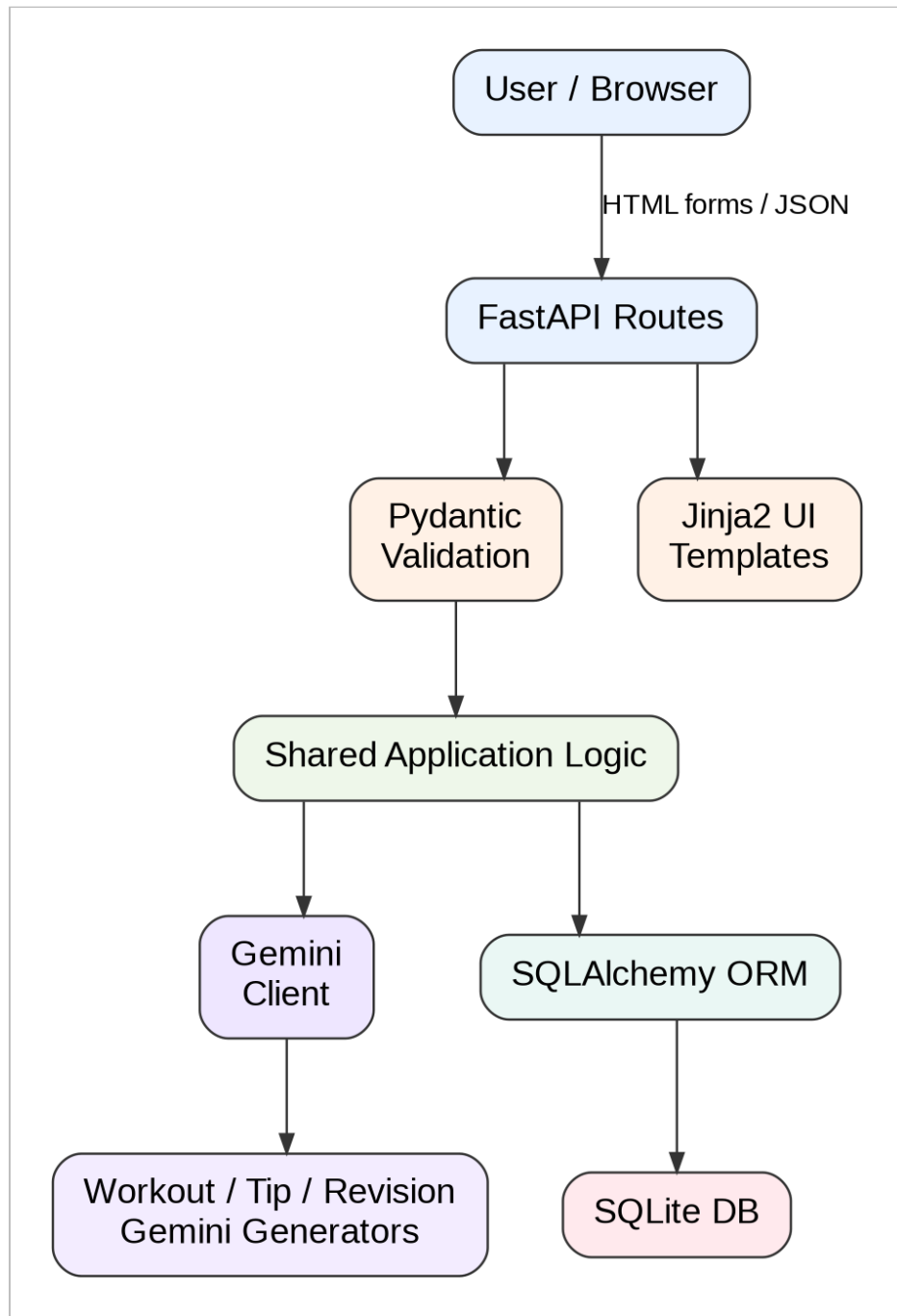


Solution Architecture

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	4 Marks



Solution Architecture Diagram

Architectural components

Component	Responsibility
<code>main.py</code>	Creates FastAPI app, mounts static files and initializes database
<code>routes.py</code>	HTML routes, shared logic and JSON API
<code>schemas.py</code>	Input/output validation contracts
<code>database.py</code>	SQLAlchemy models and persistence helpers

Component	Responsibility
<code>gemini_client.py</code>	Shared AI client, retries, fallbacks and errors
Generator modules	Workout, nutrition and revised-plan generation
Templates/CSS	Current presentation layer